TURNKEY LINUX ACTIVE DIRECTORY 18.1 DOMAIN CONTROLLER & DEBIAN CLIENT CONFIGURATION GUIDE

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LINUX CLIENT CONFIGURATION

Install Required Packages

sudo apt install -y realmd sssd sssd-tools libnss-sss libpam-sss krb5-user adcli samba-common-bin oddjob oddjob-mkhomedir

Enable Make Home Directory For AD User On First Login

sudo pam-auth-update –enable mkhomedir

Download A Copy Of The OpenRSAT Application

OpenRSAT is a very good AD management tool. Because it is GTK2 it can have some weird rendering characteristics. Often position your cursor on the window title bar and F5 to reload.

https://github.com/tranquilit/OpenRSAT

Add / Edit /etc/krb5.conf On The Linux Client & chmod 644

Set DOMAIN.LAN to match your domain name:

```
[libdefaults]
   default_realm = DOMAIN.LAN
   dns_lookup_realm = false
   dns_lookup_kdc = false

[realms]
   DOMAIN.LAN = {
      kdc = dc1.domain.lan
      admin_server = dc1.domain.lan
   }
[domain_realm]
   .domain.lan = DOMAIN.LAN
   domain.lan = DOMAIN.LAN
```

Add AD Server IP To /etc/hosts - DNS Is NOT Actually Required!

Add a line to resolve your server name to /etc/hosts:

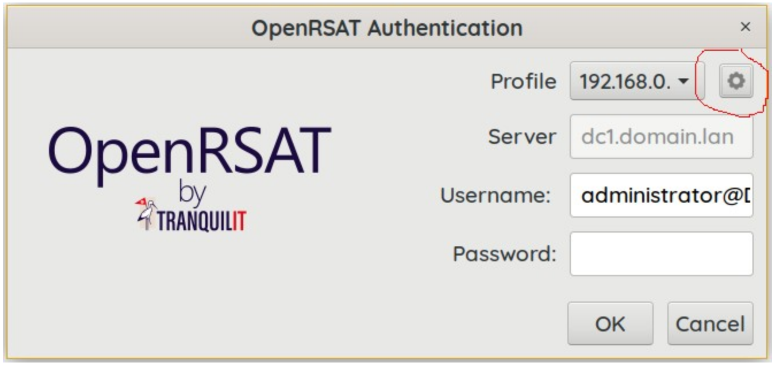
192.168.0.200 dc1 dc1.domain.lan domain.lan

Join The Domain

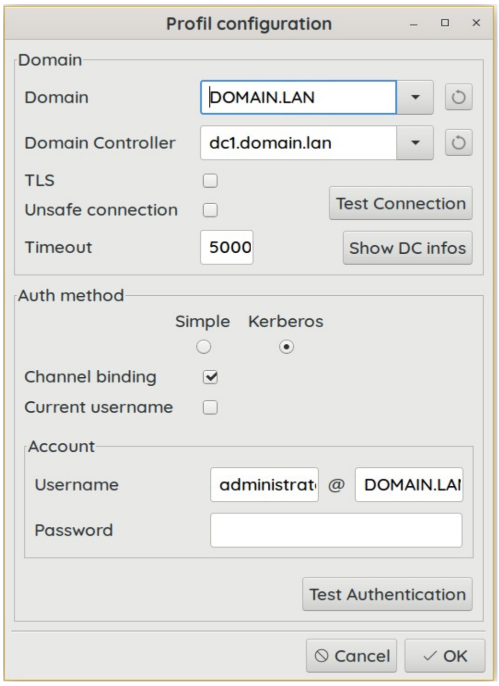
sudo realm join -U administrator domain.lan

Configure OpenRSAT To Connect To The AD Controller

Click the gear icon to add a new profile:



Add the connection parameters:



Create An AD User & Group

Use OpenRSAT, webmin: https://xxx.xxx.xxx.xxx:12321 samba module, or samba-tool in the ssh/terminal.

- On the AD server create a new group “linuxlogins@domain.lan”
- Create domain login users and add them to the group.

Give Permission To The AD Group To Login To The Linux Client

sudo realm permit -g “linuxlogins@domain.lan”

How To Leave The Domain

sudo realm leave —remove –U administrator domain.lan

There could be times you want to flush SSSD cached credentials to be sure - using these commands:

```
# sudo rm -rf /var/lib/sss/db/*
# sudo systemctl restart sssd
# sudo rm -f /etc/sssd/sssd.conf
# sudo systemctl restart realmd
```

SERVER CONFIGURATION

Configure Winbind So AD Can Be Referenced By chown:chrgp

Edit file /etc/nsswitch.conf and add winbind:

```
passwd:     files winbind
group:      files winbind
```

Install the libnss-libwind package:

```
# sudo apt-get install libnss-winbind
# ln -sf /usr/lib/x86_64-linux-gnu/libnss_winbind.so.2 /lib/x86_64-linux-gnu/libnss_winbind.so
# ldconfig
# systemctl restart winbind
# systemctl restart samba-ad-dc
```

Turnkey Default Shipped Samba Share Configuration (smb.conf)

This is the out-of-the-box Turnkey configuration (if you ever want to revert & start over):

BTW … this uses NTLM weak security, and does not enforce encryption.
Any client at all on the network can authenticate using Network Neighbourhood to locate the shares, simply to be prompted for WORKGROUP credentials.
Domain membership is not enforced in this configuration.

```
[global]

        dns forwarder = 8.8.8.8
        interfaces = 127.0.0.1 192.168.0.200 2001:8003:2762:ee00:2ed0:5aff:fe17:3365
        netbios name = DC1
        realm = DOMAIN.LAN
        server role = active directory domain controller
        workgroup = DOMAIN
        idmap_ldb:use rfc2307 = yes
```

Same with smb3 minumum protocol required and force encyption added:

```
[global]

        dns forwarder = 8.8.8.8
        interfaces = 127.0.0.1 192.168.0.200 2001:8003:2762:ee00:2ed0:5aff:fe17:3365
        netbios name = DC1
        realm = DOMAIN.LAN
        server role = active directory domain controller
        workgroup = DOMAIN
        idmap_ldb:use rfc2307 = yes
        server smb encrypt = yes
        server min protocol = SMB3
```

AD Secure Kerberos Authenticated Samba With SMB3 & Encryption

Domain Membership & AD Login Is Enforced In This Configuration.
This is probably what you want in an enterprise scenario:

```
[global]
        dns forwarder = 8.8.8.8
        interfaces = 127.0.0.1 192.168.0.200
        netbios name = DC1
        realm = DOMAIN.LAN
        server role = active directory domain controller
        workgroup = DOMAIN
        idmap_ldb:use rfc2307 = yes
        map to guest = Never
        restrict anonymous = 2
        server smb encrypt = yes
        server min protocol = SMB3
        ntlm auth = disabled
        client use kerberos = required
```

CONFIGURING A HOME SHARE & PERMISSIONS

Create the home share folder on the linux AD server:

```
# mkdir -p /srv/share/home

# chown nobody:'DOMAIN\domain users' /srv/share/home

# chmod 050 /srv/share/home
```

Add this share defintion to /etc/smb/conf:

```
[home]
   comment = Active Directory User Home Folders
   path = /srv/share/home/
   writeable = yes
   valid users = @"domain users"
   map archive = no
   map system = no
   map hidden = no
   inherit permissions = yes
   create mask = 0660
   directory mask = 0770
   force create mode = 0660
   force directory mode = 0770
   inherit owner = yes
   vfs objects = acl_xattr
   map acl inherit = yes
   store dos attributes = yes
   guest ok = no
```

Create a home sub-folder per user and restrict access to the user only:

```
# mkdir /srv/share/home/username
# chown ‘DOMAIN\<AD username>’:’DOMAIN\domain users’ /srv/share/home/username
# chmod 700 /srv/share/home/<username>
```

**** *We make domain users the group on user sub-folder but chmod 700 on the user folder.*
**** *You could change that group to be a backup operators group later and set chmod 750.*

CONFIGURING A GROUP SHARE WITH SUB-FOLDERS

Create the group share folder on the linux AD server:

```
# mkdir -p /srv/share/groups
# chown nobody:'DOMAIN\domain users' /srv/share/groups
# chmod 550 /srv/share/groups
```

Add this share definition to /etc/smb/conf:

```
[groups]
   comment = Active Directory Group Folders
   path = /srv/share/groups/
   writeable = yes
   valid users = @"domain users"
   map archive = no
   map system = no
   map hidden = no
   inherit permissions = yes
   create mask = 0660
   directory mask = 0770
   force create mode = 0660
   force directory mode = 0770
   inherit owner = yes
   vfs objects = acl_xattr
   map acl inherit = yes
   store dos attributes = yes
   guest ok = no
```

Use OpenRSAT, Webmin or samba-tool to create an AD group for example “MARKETING”.

To create a sub-folder per group and restrict access to that group using chown/chmod:

```
# mkdir /srv/share/groups/marketing
# chown nobody:’DOMAIN\MANAGEMENT’ /srv/share/groups/marketing
# chmod 2770 /srv/share/group/marketing
```

Create a sub-folder for multiple groups using combination chown/chmod & Posix ACL's

```
# mkdir /srv/share/groups/transport

# chown nobody:"DOMAIN\domain users" /srv/share/groups/transport

# chmod 2700 /srv/share/groups/transport

# setfacl -m g:"DOMAIN\transport":rw /srv/share/groups/transport

# setfacl -m g:"DOMAIN\marketing":rw /srv/share/groups/transport
```

**** *You can also use ACL's to give backup operators and groups access*

Listing POSIX ACL'S

```
getfacl -d file/foldername
```

Remove an individual ACL

```
setfacl -x g:groupname file/folderpath
```

Remove all ACL'S

```
setfacl -b file/folderpath
```

To prevent users from deleting a home or group sub-folder apply an immutable hidden file:

```
# > /srv/share/home/<AD username>/.deletion_prevented
# > chattr +i /srv/share/home/<AD username>/.deletion_prevented

# > /srv/share/groups/<AD groupname>/.deletion_prevented
# > chattr +i /srv/share/groups/<AD groupname>/.deletion_prevented
```